



THE UNIVERSITY OF
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ENGG1810 COMPUTER ENGINEERING
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MAJOR ASSIGNMENT
Student Management System

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The code is divided into two separate programmes: *student_management_system.py* and *main.py*.

The primary purpose of *student_management_system.py* is to implement and initialise our classes for student, unit, and degree information, as well as to provide access to the pre-existing database. It also defines the various functions used in *main.py*, such as inserting a new data point or displaying the existing database for all three classes.

The primary purpose of *main.py* is to implement a user-friendly interface. It makes use of the functions, methods, and classes defined in *student_management_system.py*.

Student_Management_System.py

The code contains three classes: **Student**, **Unit**, and **Degree**.

In the **Student** class, an empty dictionary **student_data** is initialised. To display basic information about a student, we have implemented the **student_basic_information** method, which simply prints the relevant details using multiple **print** commands and the **self** variable. The **insert_student_information** method allows users to add student details and create new **Student** instances, ensuring that all inputs match the required data types and specifications. Additionally, a pre-existing database of student information is stored in the class variable **student_data**.

For the **Unit** class, an empty dictionary **unit_data** is initialised. The **insert_unit_information** method allows users to input details about a unit and create a new **Unit** object, again ensuring that the data types and specifications are met. A pre-existing database of unit information is stored in the **unit_data** class variable.

The **Degree** class follows the same structure, with its own **degree_data** dictionary storing degree-related information.

In summary, the **student_data**, **unit_data**, and **degree_data** dictionaries store the data for students, units, and degrees, respectively.

Main.py

main.py is the main programme from which the student management system is run.

It first imports the entire *student_management_system* module to access all class definitions, methods, and functions. A **display_menu** function is defined to prompt the user for valid input. The **main** function then assigns more convenient variable names to the imported dictionaries.

A **while** loop keeps the programme running until the user enters 7 (Exit), which breaks the loop.

- **Choice 1:** Loops through every key-value pair in the **student_data** dictionary, printing the student number (key) and student information (value).
- **Choice 2:** Performs the same operation for the **unit_data** dictionary.

- **Choice 3:** Performs the same operation for the `degree_data` dictionary.
- **Choice 4:** Calls the `insert_student_information()` method from the `Student` class and uses the number of students + 1 as the key after validating all inputs.
- **Choice 5:** Calls the `insert_unit_information()` method from the `Unit` class, following the same process as Choice 4.
- **Choice 6:** Calls the appropriate method from the `Degree` class, following the same process as Choices 4 and 5.

Finally, the statement `if __name__ == '__main__':` ensures that the `main()` function is only executed when *main.py* is run directly, preventing any unintended execution of code from the imported module.